Lecture 10

Module 7: Storage

Module 8: Databases

Module 7: Storage

AWS Academy Cloud Foundations

Module overview

Topics

- Amazon Elastic Block Store (Amazon EBS)
- Amazon Simple Storage Service (Amazon S3)
- Amazon Elastic File System (Amazon EFS)
- Amazon Simple Storage Service Glacier

Demos

- Amazon EBS console
- Amazon S3 console
- Amazon EFS console
- Amazon S3 Glacier console

Lab

- Working with Amazon EBS

Activities

- Storage solution case study

Knowledge check

Module objectives

After completing this module, you should be able to:

- Identify the different types of storage
- Explain Amazon S3
- Identify the functionality in Amazon S3
- Explain Amazon EBS
- Identify the functionality in Amazon EBS
- Perform functions in Amazon EBS to build an Amazon EC2 storage solution
- Explain Amazon EFS
- Identify the functionality in Amazon EFS
- Explain Amazon S3 Glacier
- Identify the functionality in Amazon S3 Glacier
- Differentiate between Amazon EBS, Amazon S3, Amazon EFS, and Amazon S3 Glacier

Core AWS services

Amazon Virtual Private Cloud (Amazon VPC)

Amazon Elastic Compute Cloud (Amazon EC2)

Amazon S3

Amazon EBS

Amazon EFS

Amazon S3 Glacier

Storage

Amazon Relational Database Service

Amazon DynamoDB

Database

AWS Identity and Access Management (IAM)

Section 1: Amazon Elastic Block Store (Amazon EBS)

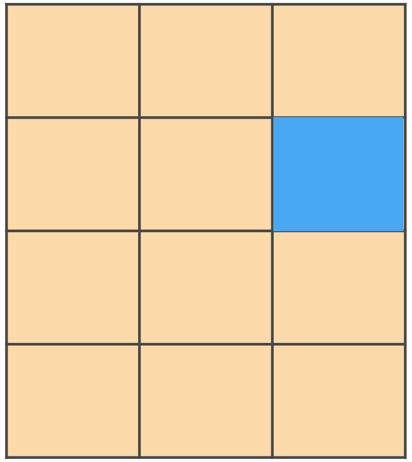
Module 7: Storage

Storage

Amazon Elastic Block Store
(Amazon EBS)

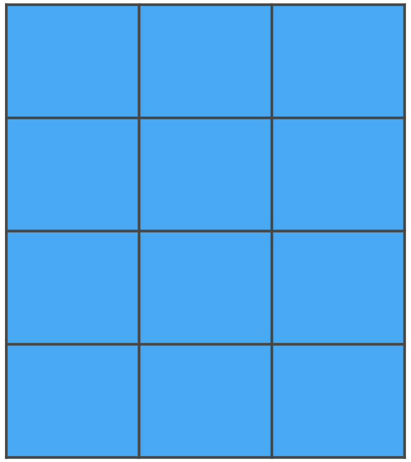
AWS storage options: Block storage versus object storage

What if you want to change **one character** in a 1-GB file?



Block storage

Change one block (piece of the file) that contains the character



Object storage

Entire file must be updated

Amazon EBS

Amazon EBS enables you to **create individual storage volumes** and **attach them** to an Amazon EC2 instance:

- Amazon EBS offers block-level storage.
- Volumes are automatically replicated within its Availability Zone.
- It can be backed up automatically to Amazon S3 through snapshots.
- Uses include –
  - Boot volumes and storage for Amazon Elastic Compute Cloud (Amazon EC2) instances
  - Data storage with a file system
  - Database hosts
  - Enterprise applications

Amazon EBS volume types

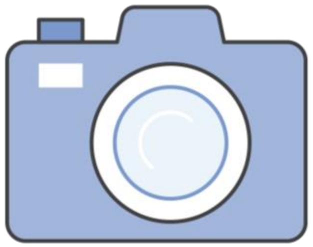
	Solid State Drives (SSD)		Hard Disk Drives (HDD)	
	General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
Maximum Volume Size	16 TiB	16 TiB	16 TiB	16 TiB
Maximum IOPS/Volume	16,000	64,000	500	250
Maximum Throughput/Volume	250 MiB/s	1,000 MiB/s	500 MiB/s	250 MiB/s

Amazon EBS volume type use cases

Solid State Drives (SSD)		Hard Disk Drives (HDD)	
General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
• This type is recommended for most workloads	• Critical business applications that require sustained IOPS performance, or more than 16,000 IOPS or 250 MiB/second of throughput per volume	• Streaming workloads that require consistent, fast throughput at a low price	• Throughput-oriented storage for large volumes of data that is infrequently accessed
• System boot volumes	• Large database workloads	• Big data	• Scenarios where the lowest storage cost is important
• Virtual desktops		• Data warehouses	• It cannot be a boot volume
• Low-latency interactive applications		• Log processing	
• Development and test environments		• It cannot be a boot volume	

Amazon EBS features

- Snapshots –
  - Point-in-time snapshots
  - Recreate a new volume at any time



- Encryption –
  - Encrypted Amazon EBS volumes
  - No additional cost

- Elasticity –
  - Increase capacity
  - Change to different types

Amazon EBS: Volumes, IOPS, and pricing

1. **Volumes –**
   - Amazon EBS volumes persist independently from the instance.
   - All volume types are charged by the amount that is provisioned per month.
2. **IOPS –**
   - General Purpose SSD:
     - Charged by the amount that you provision in GB per month until storage is released.
   - Magnetic:
     - Charged by the number of requests to the volume.
   - Provisioned IOPS SSD:
     - Charged by the amount that you provision in IOPS (multiplied by the percentage of days that you provision for the month).

Amazon EBS: Snapshots and data transfer

3. Snapshots –

- Added cost of Amazon EBS snapshots to Amazon S3 is per GB-month of data stored.

4. Data transfer –

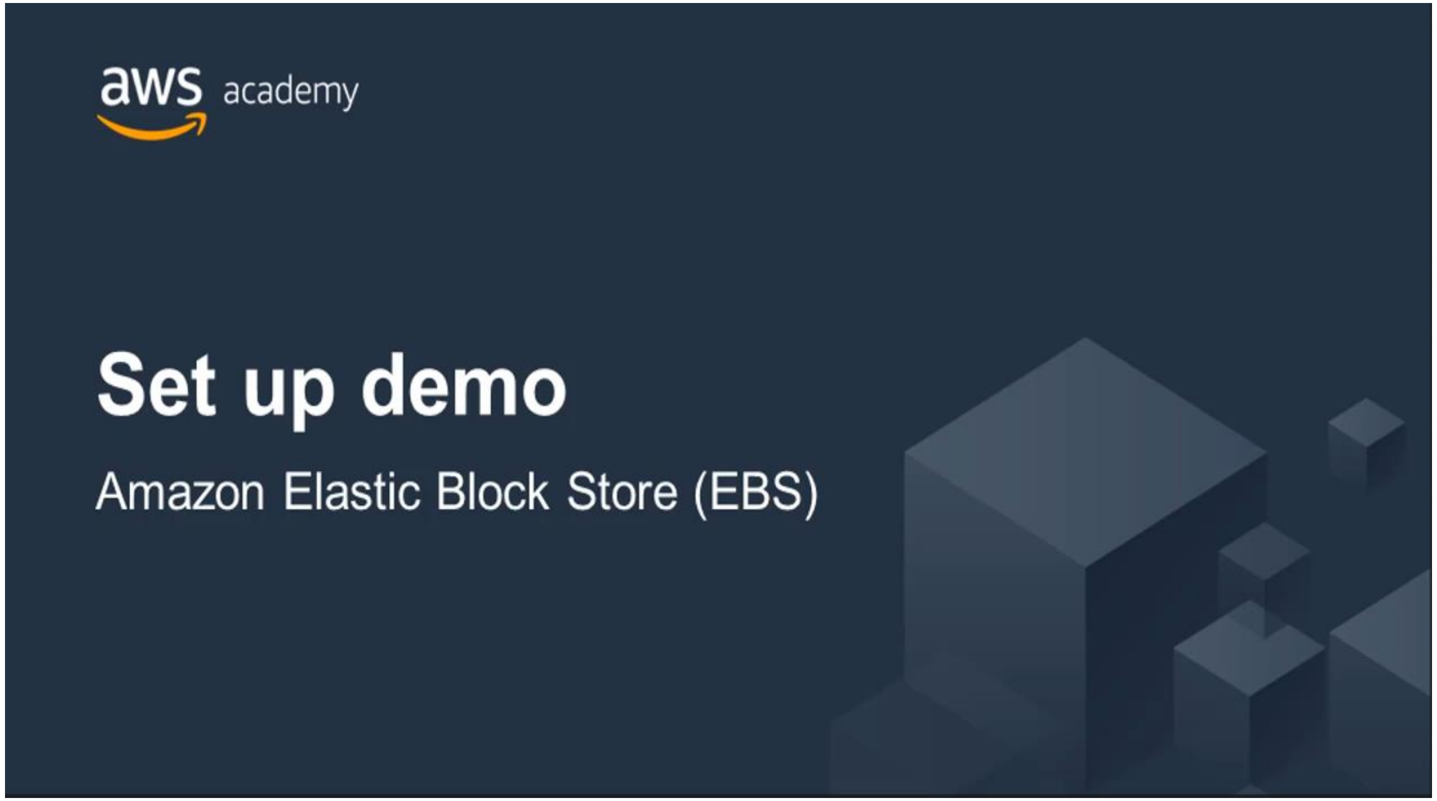
- Inbound data transfer is free.
- Outbound data transfer across Regions incurs charges.

Section 1 key takeaways

Amazon EBS features:

- Persistent and customizable block storage for Amazon EC2
- HDD and SSD types
- Replicated in the same Availability Zone
- Easy and transparent encryption
- Elastic volumes
- Back up by using snapshots

Recorded demo: Amazon Elastic Block Store



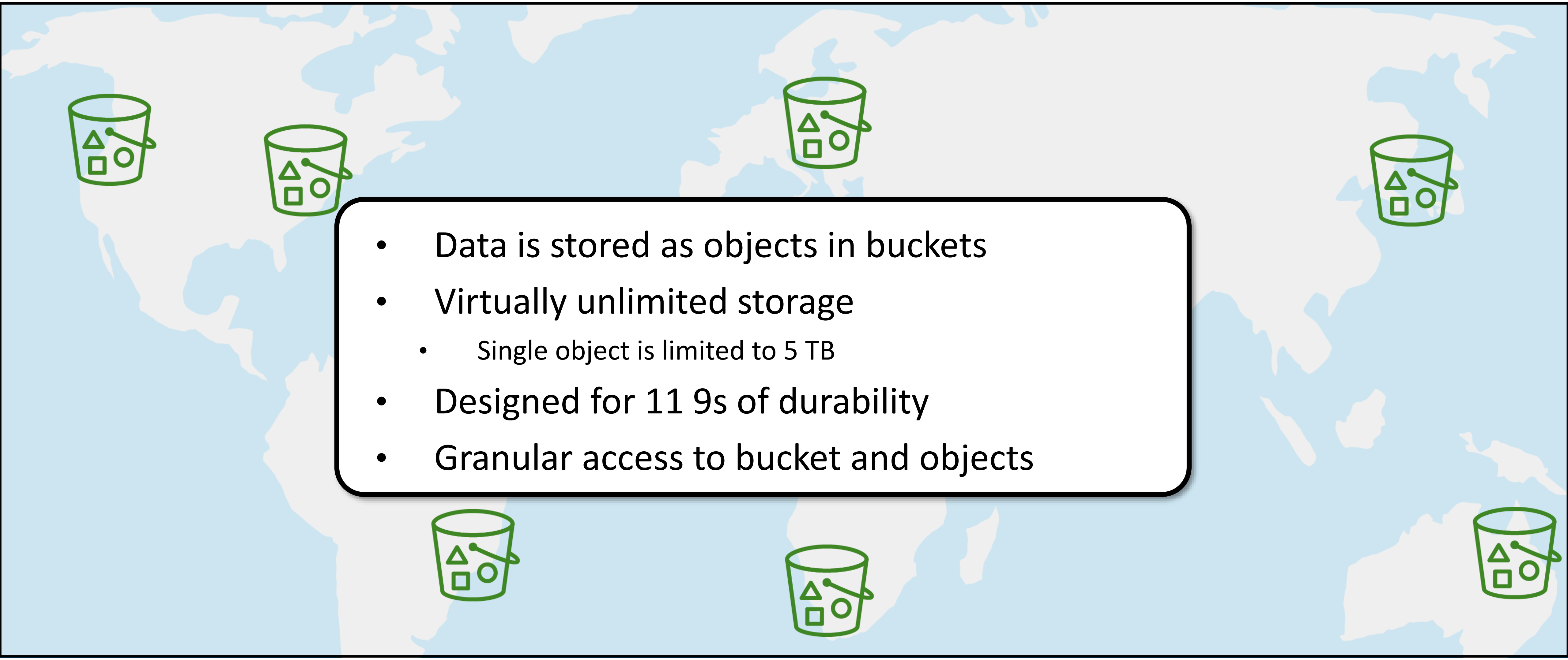
Section 2: Amazon Simple Storage Service (Amazon S3)

Module 7: Storage

Storage

Amazon Simple Storage Service
(Amazon S3)

Amazon S3 overview



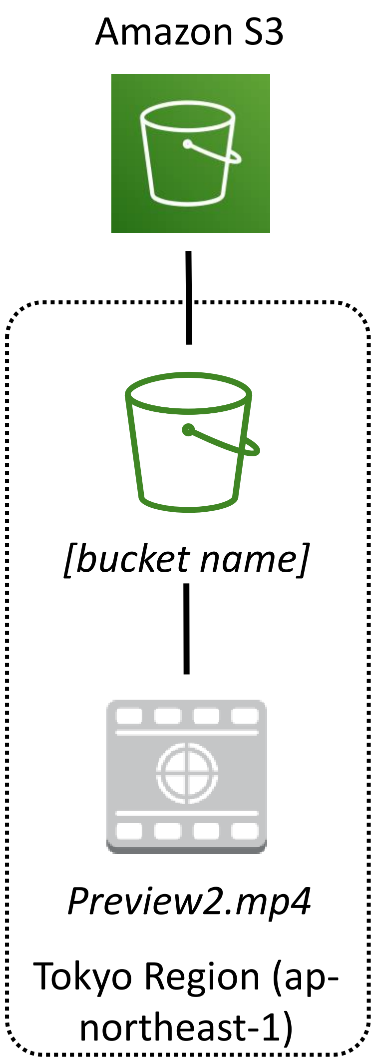
- Data is stored as objects in buckets
- Virtually unlimited storage
  - Single object is limited to 5 TB
- Designed for 11 9s of durability
- Granular access to bucket and objects

Amazon S3 storage classes

Amazon S3 offers a range of object-level storage classes that are designed for different use cases:

- Amazon S3 Standard
- Amazon S3 Intelligent-Tiering
- Amazon S3 Standard-Infrequent Access (Amazon S3 Standard-IA)
- Amazon S3 One Zone-Infrequent Access (Amazon S3 One Zone-IA)
- Amazon S3 Glacier
- Amazon S3 Glacier Deep Archive

Amazon S3 bucket URLs (two styles)



To upload your data:

1. Create a **bucket** in an AWS Region.
2. Upload almost any number of **objects** to the bucket.

Bucket path-style URL endpoint:

https://s3.ap-northeast-1.amazonaws.com/bucket-name

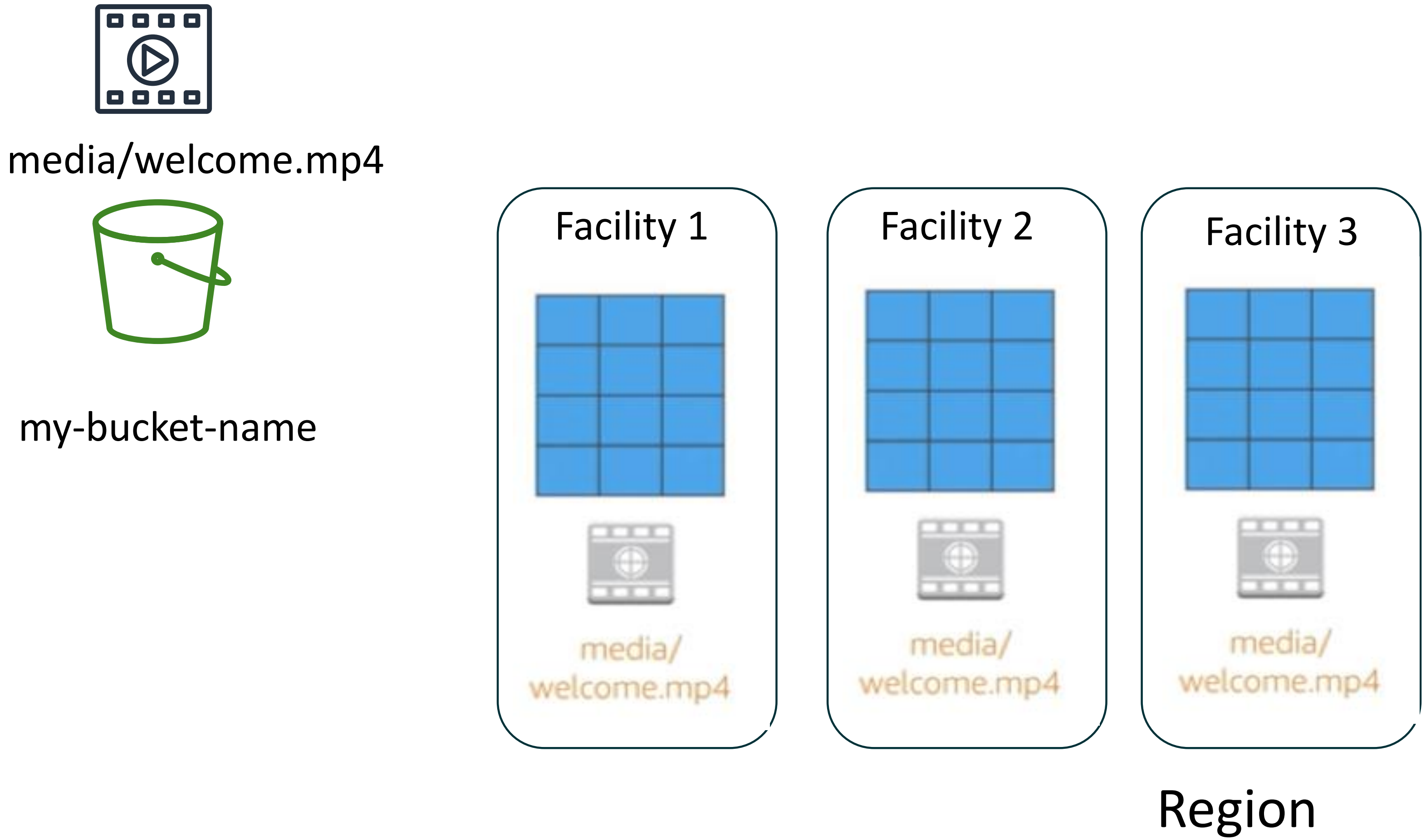
Region code (ap-northeast-1) — **Bucket name** (bucket-name)

Bucket virtual hosted-style URL endpoint:

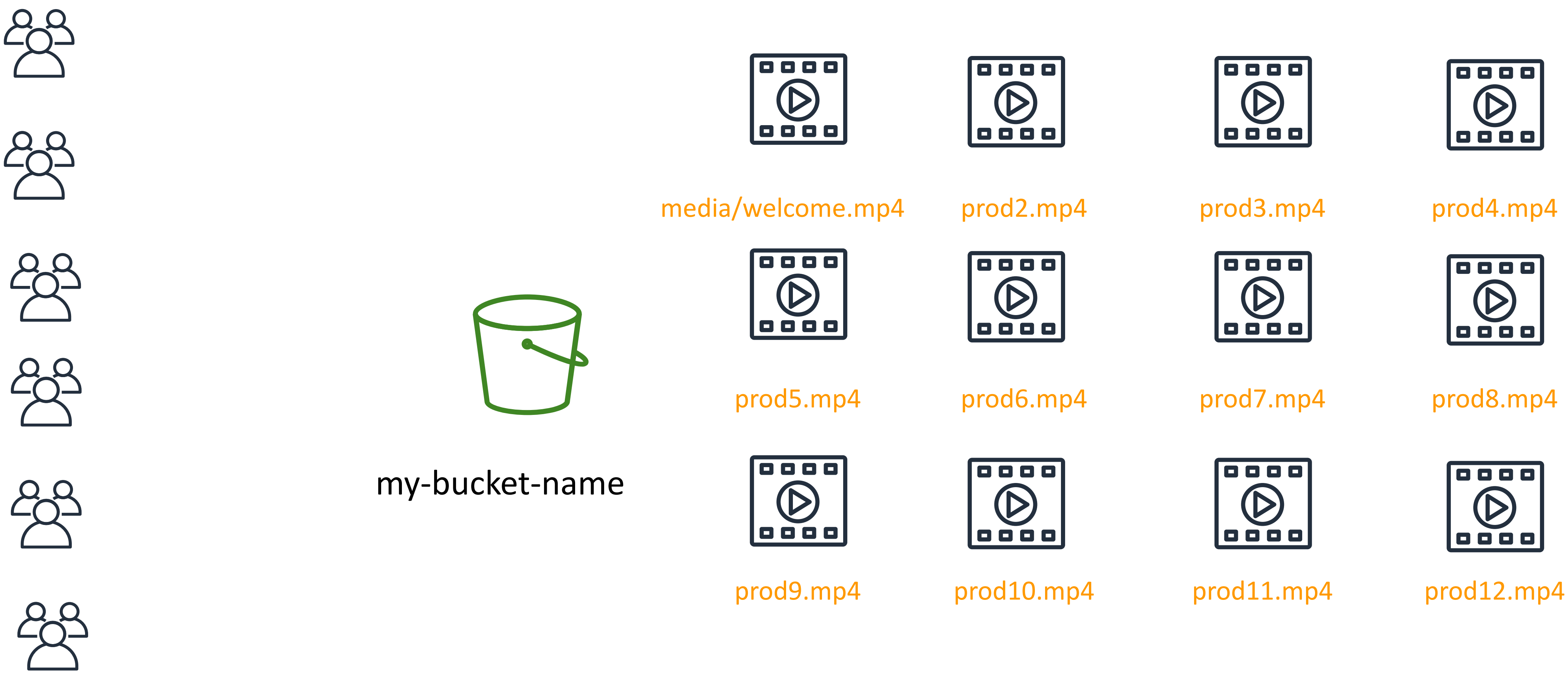
https:// bucket-name.s3-ap-northeast-1.amazonaws.com

Bucket name (bucket-name) — **Region code** (ap-northeast-1)

Data is redundantly stored in the Region



Designed for seamless scaling



Access the data anywhere

AWS Management Console

AWS Command Line Interface

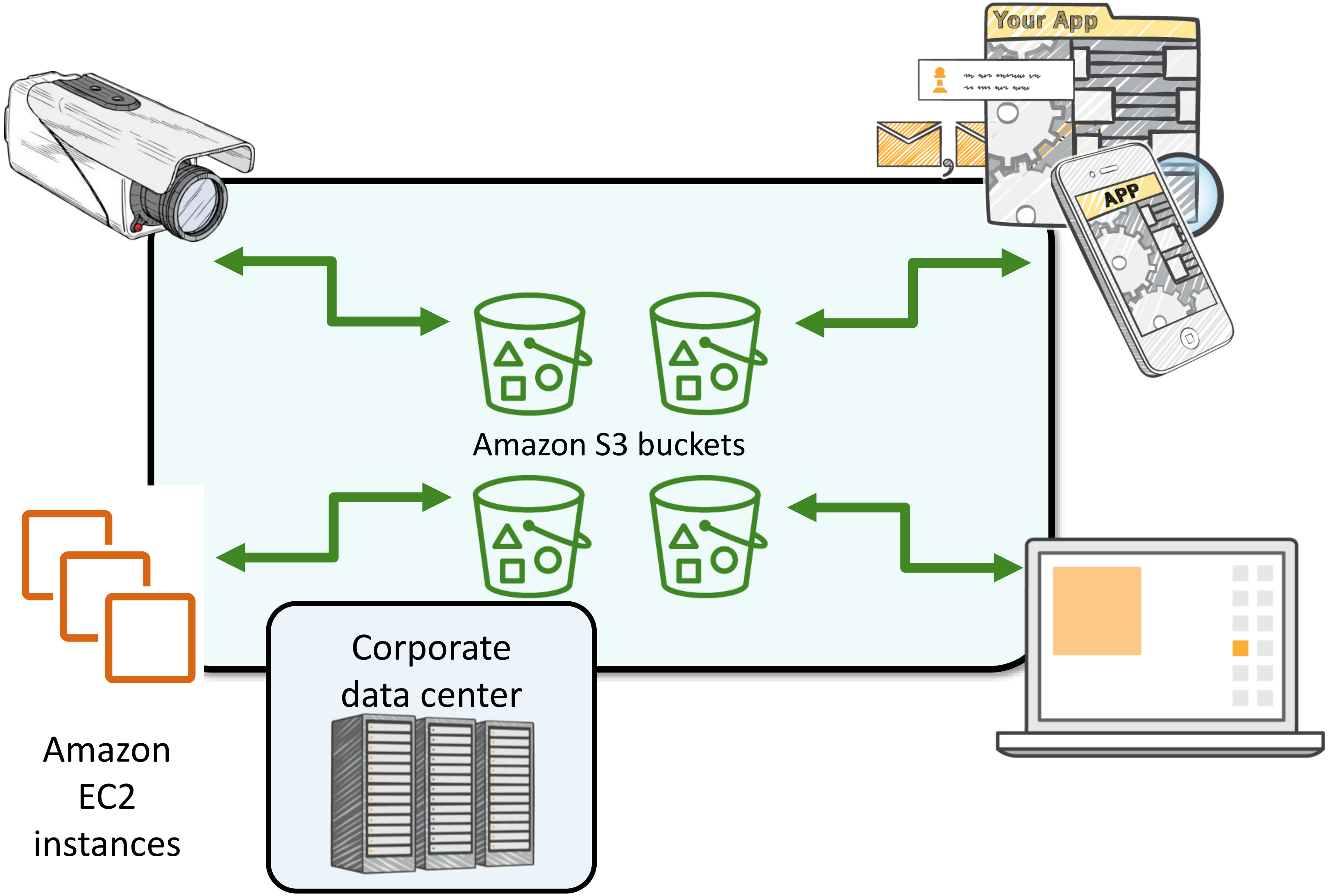
SDK

Common use cases

- Storing application assets
- Static web hosting
- Backup and disaster recovery (DR)
- Staging area for big data
- *Many more….*

Amazon S3 common scenarios

- Backup and storage
- Application hosting
- Media hosting
- Software delivery



Amazon S3 pricing

- Pay only for what you use, including –
  - GBs per month
  - Transfer OUT to other Regions
  - PUT, COPY, POST, LIST, and GET requests

- You do not pay for –
  - Transfers IN to Amazon S3
  - Transfers OUT from Amazon S3 to Amazon CloudFront or Amazon EC2 in the same Region

Amazon S3: Storage pricing (1 of 2)

To estimate Amazon S3 costs, consider the following:

1. Storage class type –
   - Standard storage is designed for:
     - 11 9s of durability
     - Four 9s of availability
   - S3 Standard-Infrequent Access (S-IA) is designed for:
     - 11 9s of durability
     - Three 9s of availability
2. Amount of storage –
   - The number and size of objects

Amazon S3: Storage pricing (2 of 2)

3. Requests –
    - The number and type of requests **(GET, PUT, COPY)**
    - Type of requests:
        - Different rates for GET requests than other requests.
4. Data transfer –
    - Pricing is based on the amount of data that is transferred out of the Amazon S3 Region
        - Data transfer in is free, but you incur charges for data that is transferred out.

Section 2 key takeaways

- Amazon S3 is a fully managed cloud storage service.
- You can store a virtually unlimited number of objects.
- You pay for only what you use.
- You can access Amazon S3 at any time from anywhere through a URL.
- Amazon S3 offers rich security controls.

Recorded demo: Amazon Simple Storage System

Section 3: Amazon Elastic File System (Amazon EFS)

Module 7: Storage

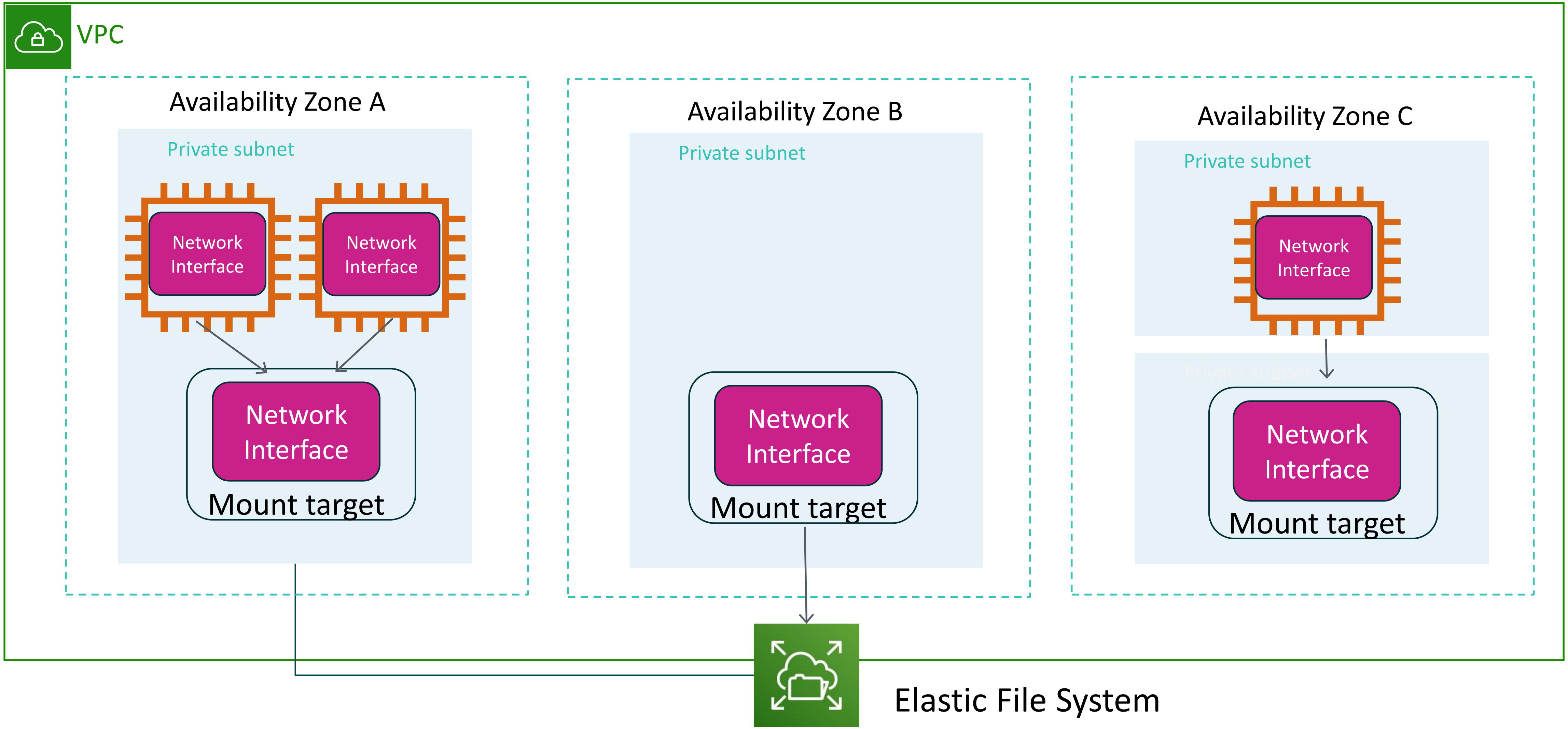
Storage

Amazon Elastic File System
(Amazon EFS)

Amazon EFS features

- File storage in the AWS Cloud
- Works well for big data and analytics, media processing workflows, content management, web serving, and home directories
- Petabyte-scale, low-latency file system
- Shared storage
- Elastic capacity
- Supports Network File System (NFS) versions 4.0 and 4.1 (NFSv4)
- Compatible with all Linux-based AMIs for Amazon EC2

Amazon EFS architecture



Amazon EFS implementation

1. Create your Amazon EC2 resources and launch your Amazon EC2 instance.
2. Create your Amazon EFS file system.
3. Create your mount targets in the appropriate subnets.
4. Connect your Amazon EC2 instances to the mount targets.
5. Verify the resources and protection of your AWS account.

Amazon EFS resources

File system

- Mount target
  - Subnet ID
  - Security groups
  - One or more per file system
  - Create in a VPC subnet
  - One per Availability Zone
  - Must be in the same VPC
- Tags
  - Key-value pairs

Section 3 key takeaways

- Amazon EFS provides file storage over a network.
- Perfect for big data and analytics, media processing workflows, content management, web serving, and home directories.
- Fully managed service that eliminates storage administration tasks.
- Accessible from the console, an API, or the CLI.
- Scales up or down as files are added or removed and you pay for what you use.

Recorded demo: Amazon Elastic File System

Section 4: Amazon S3 Glacier

Module 7: Storage

Storage

Amazon S3 Glacier

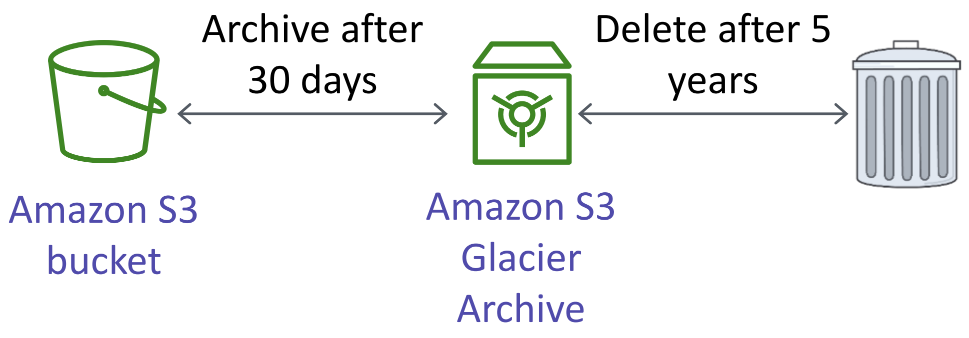
Amazon S3 Glacier review

Amazon S3 Glacier is a data archiving service that is designed for security, durability, and an extremely low cost.

- Amazon S3 Glacier is designed to provide 11 9s of durability for objects.
- It supports the encryption of data in transit and at rest through Secure Sockets Layer (SSL) or Transport Layer Security (TLS).
- The Vault Lock feature enforces compliance through a policy.
- Extremely low-cost design works well for long-term archiving.
    - Provides three options for access to archives—expedited, standard, and bulk—retrieval times range from a few minutes to several hours.

Amazon S3 Glacier

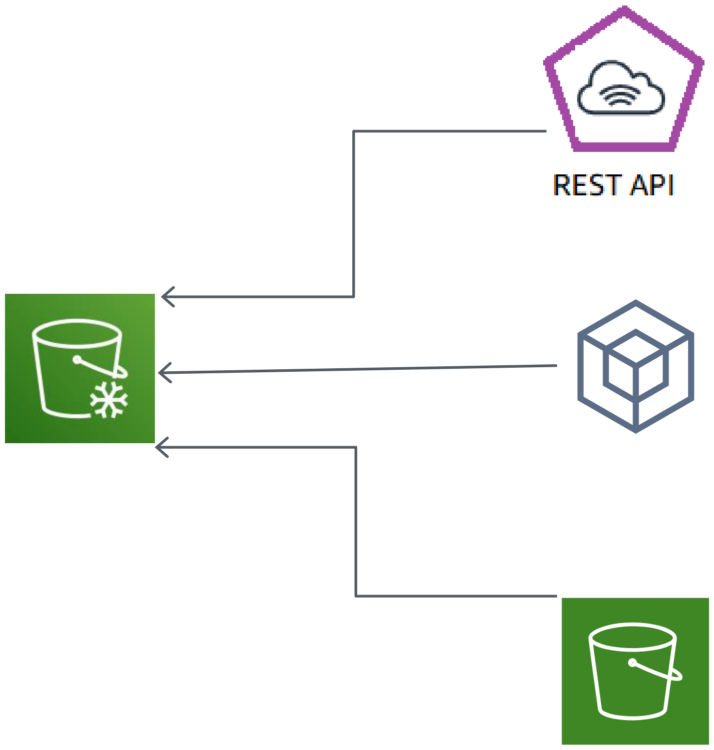
- Storage service for low-cost data archiving and long-term backup
- You can configure lifecycle archiving of Amazon S3 content to Amazon S3 Glacier
- Retrieval options –
  - Standard: 3–5 hours
  - Bulk: 5–12 hours
  - Expedited: 1–5 minutes



Amazon S3 Glacier use cases

- Media asset archiving
- Healthcare information archiving
- Regulatory and compliance archiving
- Scientific data archiving
- Digital preservation
- Magnetic tape replacement

Using Amazon S3 Glacier



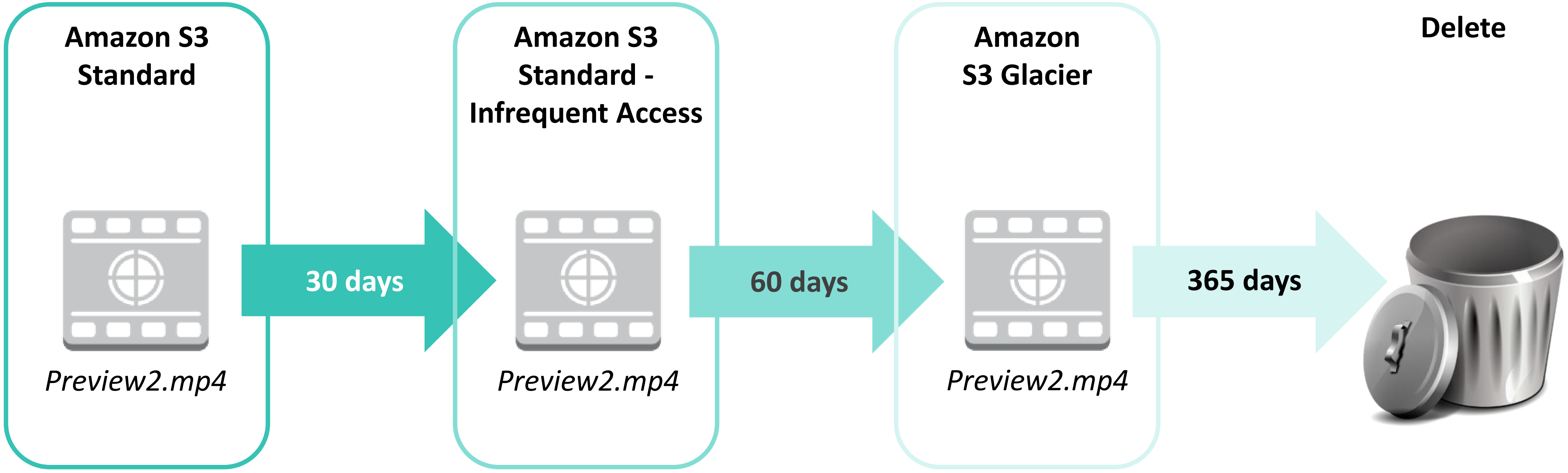
RESTful
web services

Java or .NET
SDKs

Amazon S3 with
lifecycle policies

Lifecycle policies

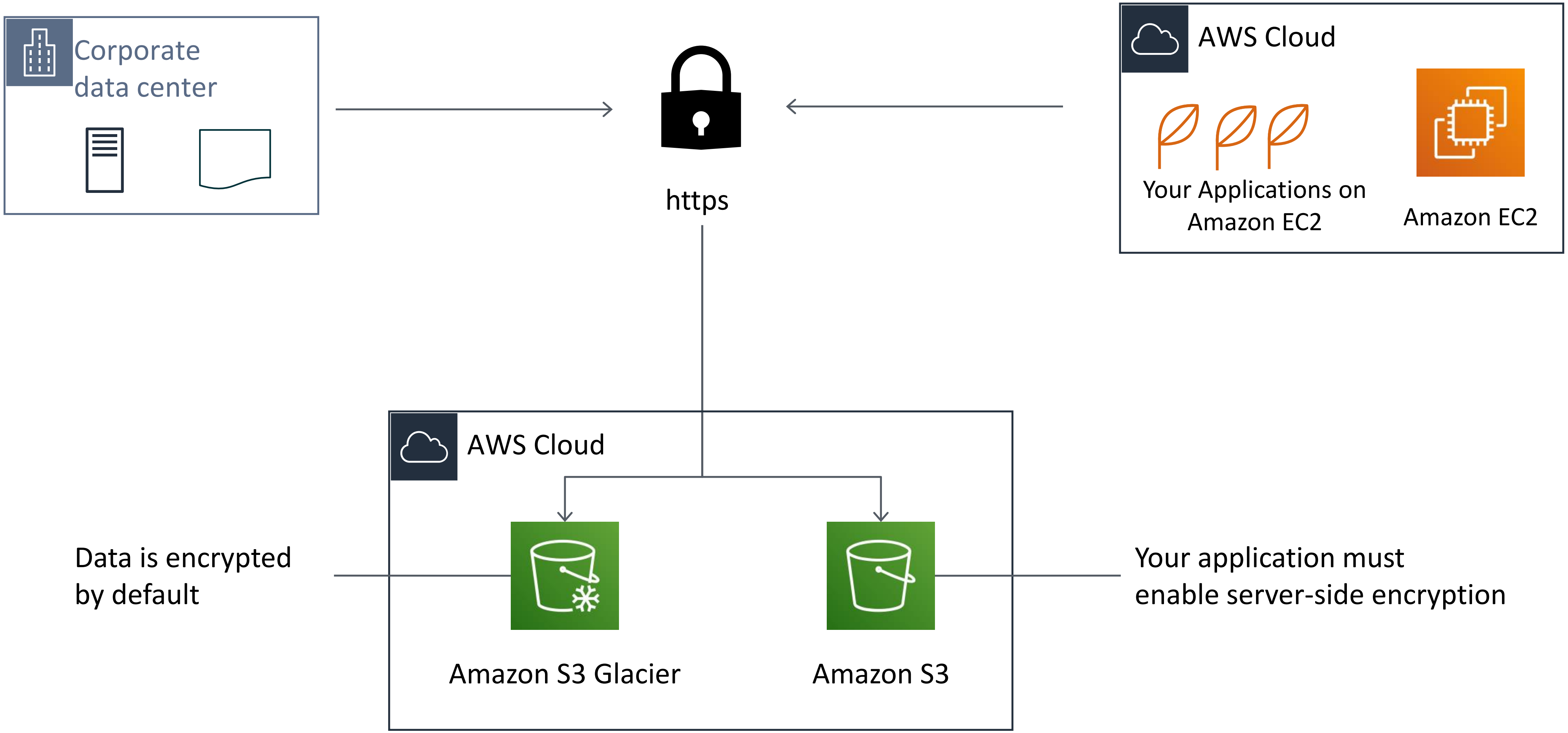
Amazon S3 lifecycle policies enable you to delete or move objects based on age.



Storage comparison

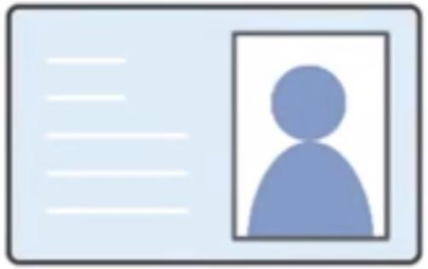
	Amazon S3	Amazon S3 Glacier
Data Volume	No limit	No limit
Average Latency	ms	minutes/hours
Item Size	5 TB maximum	40 TB maximum
Cost/GB per Month	Higher cost	Lower cost
Billed Requests	PUT, COPY, POST, LIST, and GET	UPLOAD and retrieval
Retrieval Pricing	¢ Per request	¢¢ Per request and per GB

Server-side encryption



Security with Amazon S3 Glacier

Amazon S3 Glacier



Control access with IAM

Amazon S3 Glacier encrypts your data with AES-256

Amazon S3 Glacier manages your keys for you

Section 4 key takeaways

- Amazon S3 Glacier is a data archiving service that is designed for security, durability, and an extremely low cost.
- Amazon S3 Glacier pricing is based on Region.
- Its extremely low-cost design works well for long-term archiving.
- The service is designed to provide 11 9s of durability for objects.

Recorded demo: Amazon S3 Glacier

Module wrap-up

Module 7: Storage

Module summary

In summary, in this module, you learned how to:

- Identify the different types of storage
- Explain Amazon S3
- Identify the functionality in Amazon S3
- Explain Amazon EBS
- Identify the functionality in Amazon EBS
- Perform functions in Amazon EBS to build an Amazon EC2 storage solution
- Explain Amazon EFS
- Identify the functionality in Amazon EFS
- Explain Amazon S3 Glacier
- Identify the functionality in Amazon S3 Glacier
- Differentiate between Amazon EBS, Amazon S3, Amazon EFS, and Amazon S3 Glacier

Complete the knowledge check

Sample exam question

A company wants to store data that is not frequently accessed. What is the best and cost-effective solution that should be considered?

Choice	Response
A	AWS Storage Gateway
B	Amazon Simple Storage Service Glacier
C	Amazon Elastic Block Store (Amazon EBS)
D	Amazon Simple Storage Service (Amazon S3)

Additional resources

- AWS Storage page: https://aws.amazon.com/products/storage/
- Storage Overview: https://docs.aws.amazon.com/whitepapers/latest/aws-overview/storage-services.html
- Recovering files from an Amazon EBS volume backup: https://aws.amazon.com/blogs/compute/recovering-files-from-an-amazon-ebs-volume-backup/
- Confused by AWS Storage Options? S3, EFS, EBS Explained: https://dzone.com/articles/confused-by-aws-storage-options-s3-ebs-amp-efs-explained

Module 8: Databases

AWS Academy Cloud Foundations

Module overview

Topics

- Amazon Relational Database Service (Amazon RDS)
- Amazon DynamoDB
- Amazon Redshift
- Amazon Aurora

Demos

- Amazon RDS console
- Amazon DynamoDB console

Lab

- Lab 5: Build Your DB Server and Interact with Your DB Using an App

Activity

- Database case studies

Knowledge check

Module objectives

After completing this module, you should be able to:

- Explain Amazon Relational Database Service (Amazon RDS)
- Identify the functionality in Amazon RDS
- Explain Amazon DynamoDB
- Identify the functionality in Amazon DynamoDB
- Explain Amazon Redshift
- Explain Amazon Aurora
- Perform tasks in an RDS database, such as launching, configuring, and interacting

Section 1: Amazon Relational Database Service

Module 8: Databases

Amazon Relational Database Service

Amazon Relational Database Service (Amazon RDS)

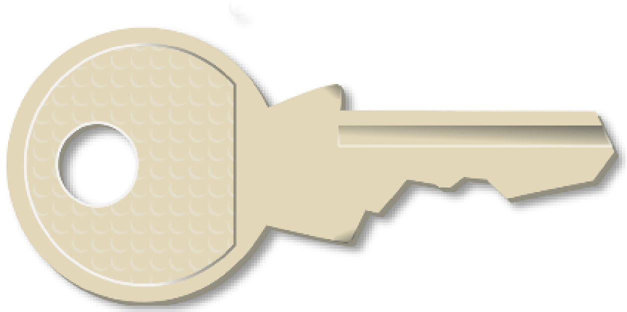
Unmanaged versus managed services

Unmanaged:

Scaling, fault tolerance, and availability are managed by you.

Managed:

Scaling, fault tolerance, and availability are typically built into the service.

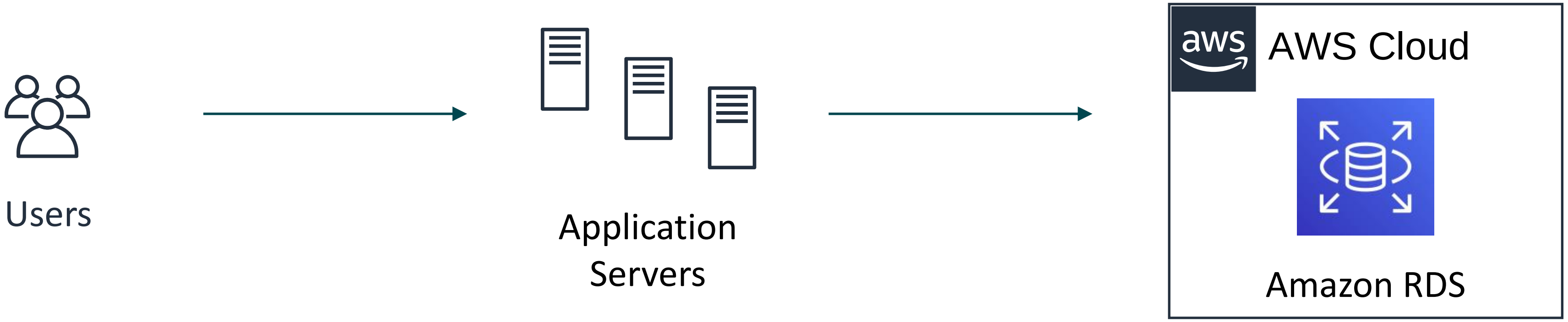


Challenges of relational databases

- Server maintenance and energy footprint
- Software installation and patches
- Database backups and high availability
- Limits on scalability
- Data security
- Operating system (OS) installation and patches

Amazon RDS

Managed service that sets up and operates a relational database in the cloud.



From on-premises databases to Amazon RDS

On-premises database → Database in Amazon Elastic Compute Cloud (Amazon EC2) → Database in Amazon RDS or Amazon Aurora

On-premises database

Application optimization
Scaling
High Availability
Database backups
Database software patches
Database software installs
Operation system patches
Operating system install
Server maintenance
Rack and stack servers
Power, HVAC, network

Database in Amazon Elastic Compute Cloud (Amazon EC2)

Application optimization
Scaling
High Availability
Database backups
Database software patches
Database software installs
Operation system patches
Operating system install
Server maintenance

AWS provides:
Rack and stack servers
Power, HVAC, network

Database in Amazon RDS or Amazon Aurora

Application optimization

AWS provides:
Scaling
High Availability
Database backups
Database software patches
Database software installs
Operation system patches
Operating system install
Server maintenance
Rack and stack servers
Power, HVAC, network

Managed services responsibilities

You manage:

- Application optimization

AWS manages:

- OS installation and patches
- Database software installation and patches
- Database backups
- High availability
- Scaling
- Power and racking and stacking servers
- Server maintenance

Amazon RDS

Amazon RDS DB instances

Amazon RDS

Amazon RDS DB main instance

DB Instance Class

- CPU
- Memory
- Network performance

DB Instance Storage

- Magnetic
- General Purpose (solid state drive, or SSD)
- Provisioned IOPS

MySQL

Amazon Aurora

Microsoft SQL Server

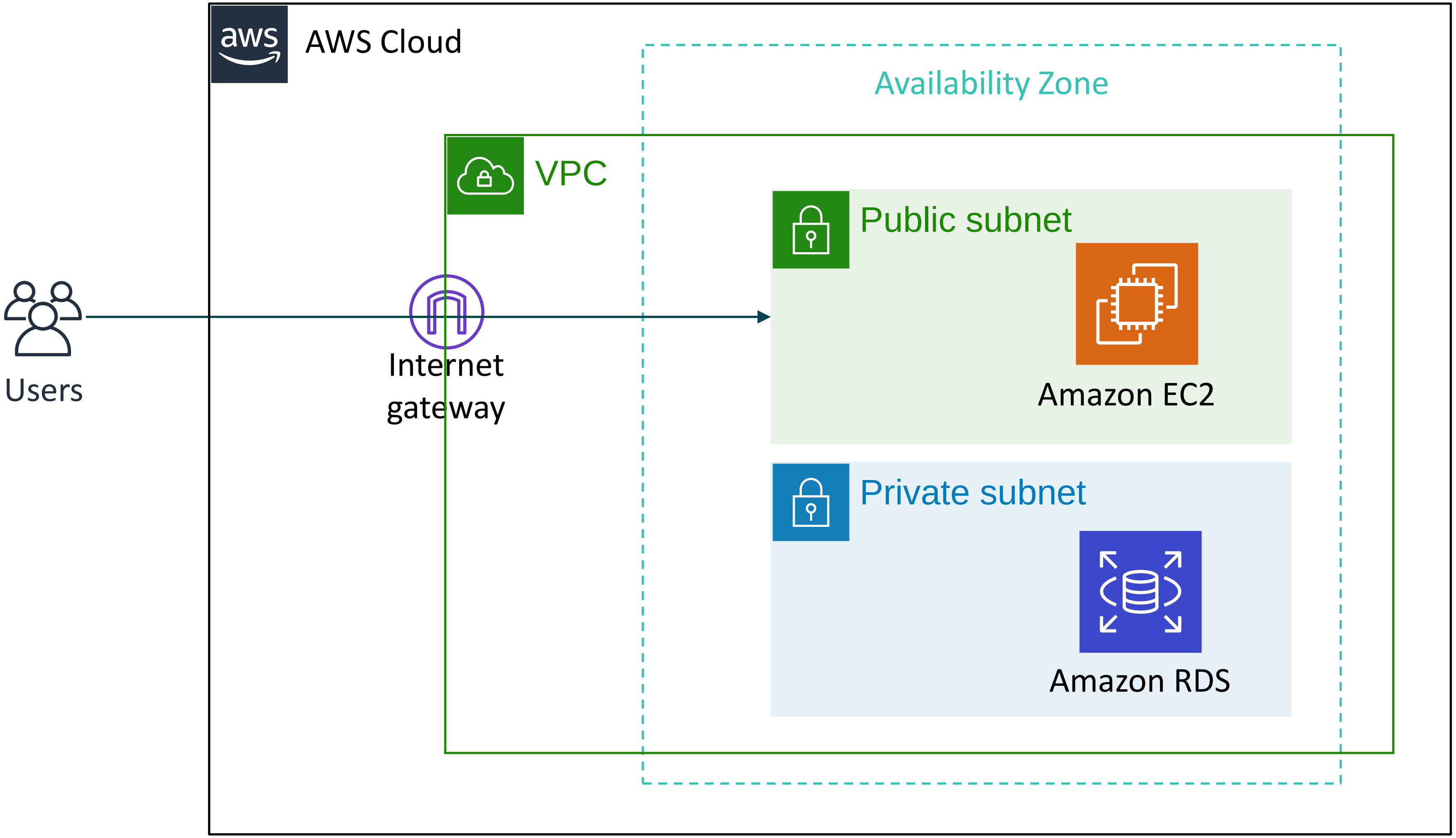
PostgreSQL

MariaDB

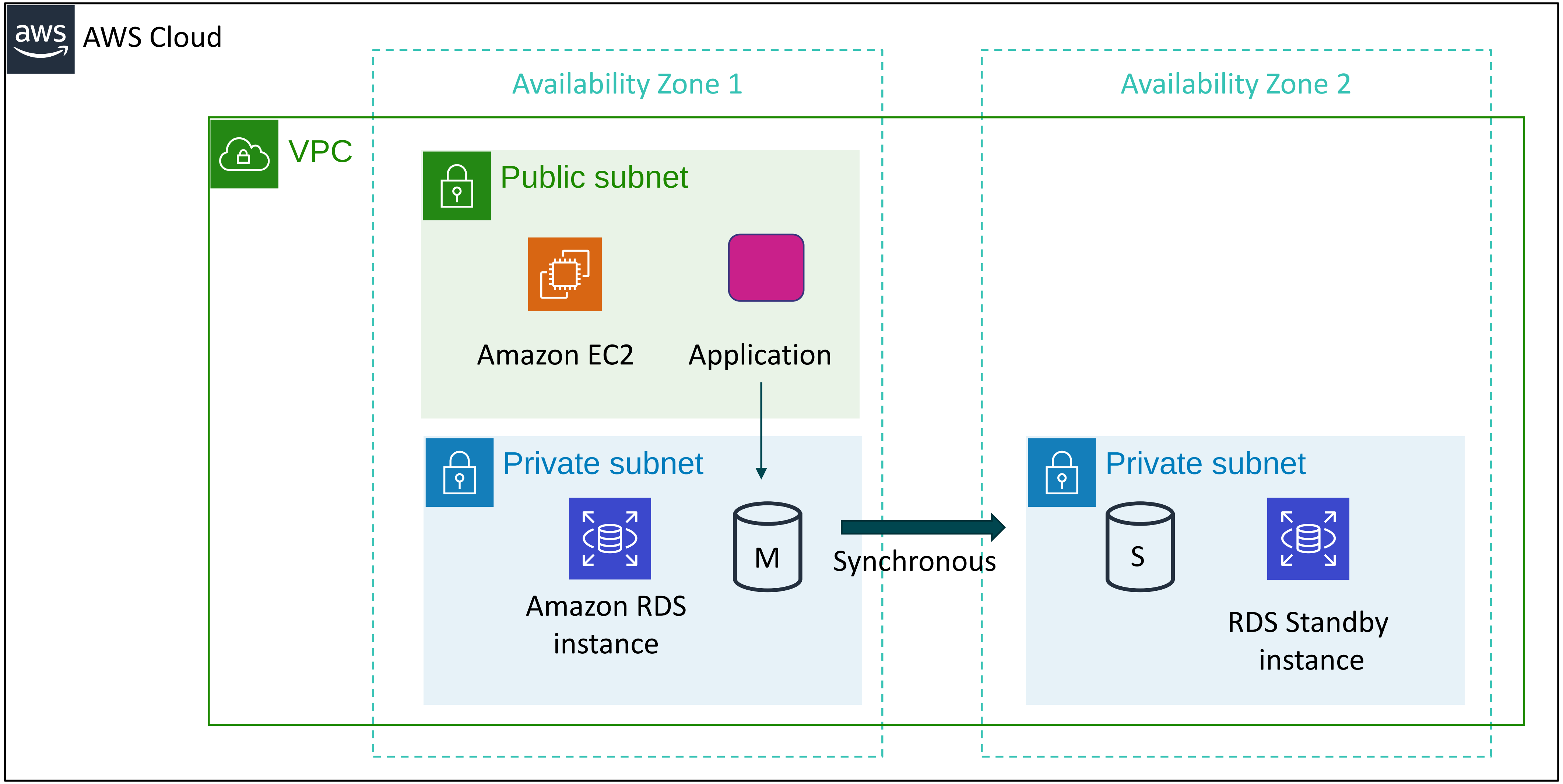
Oracle

DB engines

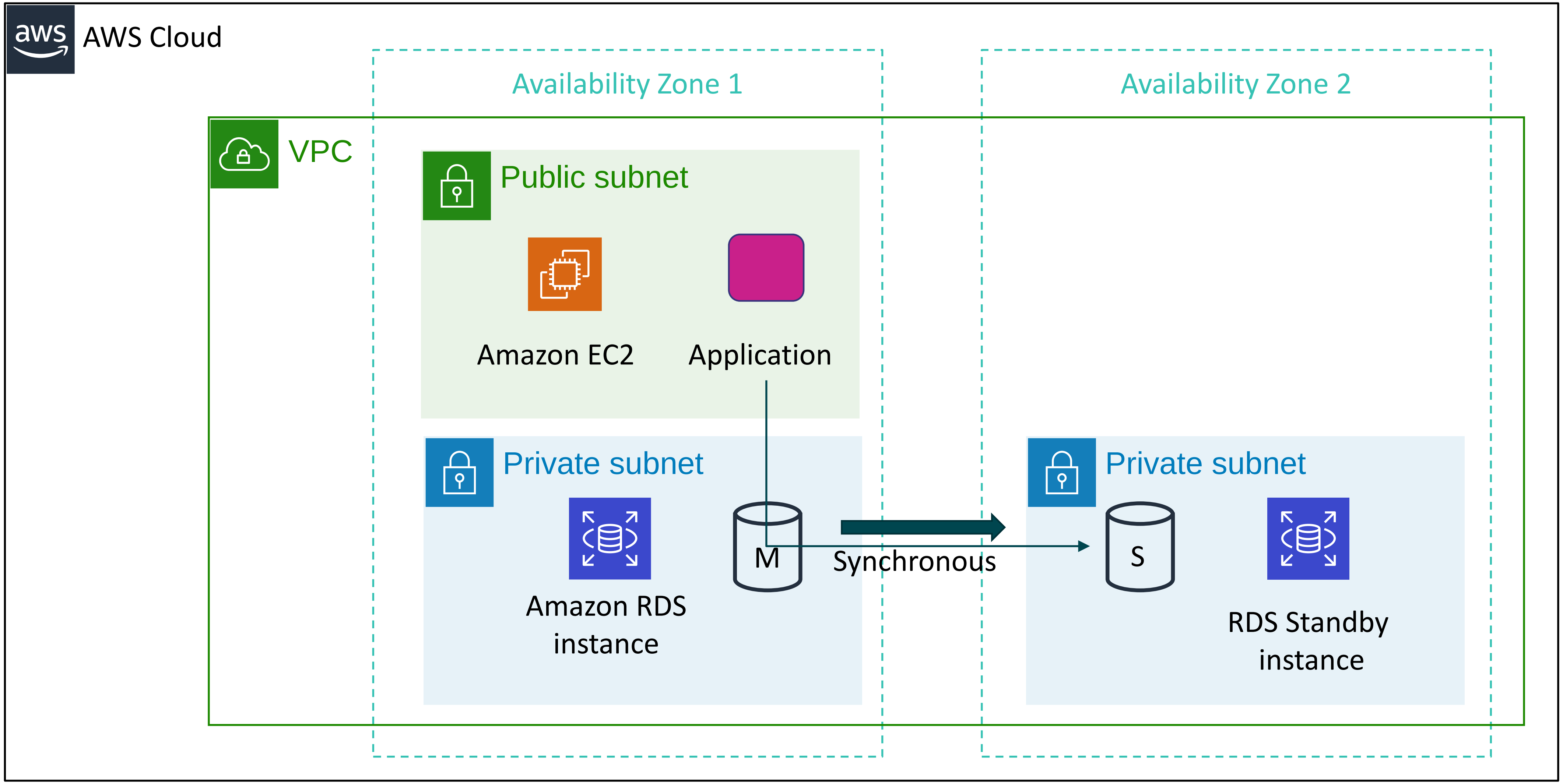
Amazon RDS in a virtual private cloud (VPC)



High availability with Multi-AZ deployment (1 of 2)



High availability with Multi-AZ deployment (2 of 2)



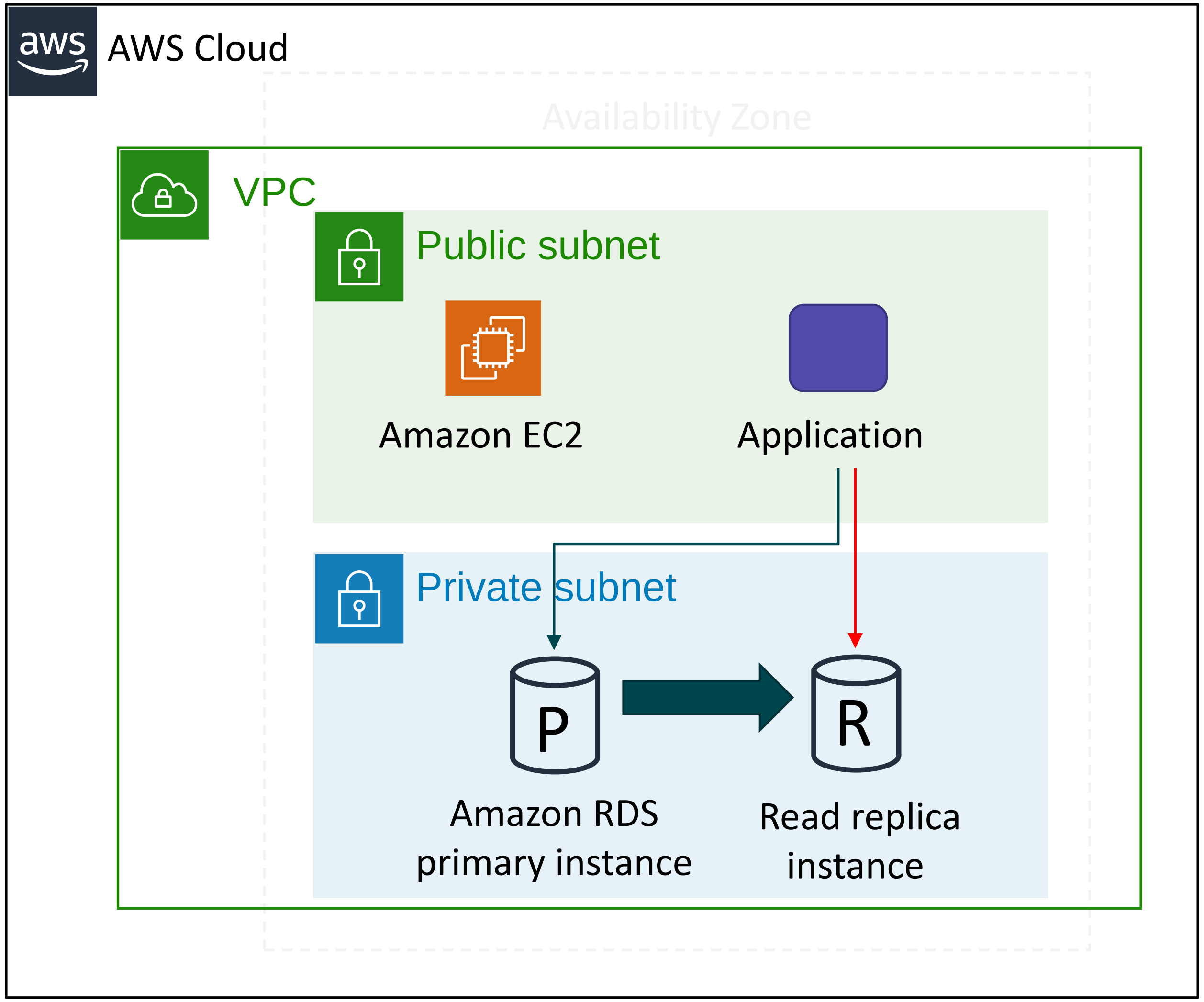
Amazon RDS read replicas

Features

- Offers asynchronous replication
- Can be promoted to primary if needed

Functionality

- Use for read-heavy database workloads
- Offload read queries



Use cases

Web and mobile applications	✓ High throughput ✓ Massive storage scalability ✓ High availability
Ecommerce applications	✓ Low-cost database ✓ Data security ✓ Fully managed solution
Mobile and online games	✓ Rapidly grow capacity ✓ Automatic scaling ✓ Database monitoring

When to Use Amazon RDS

Use Amazon RDS when your application requires:

- Complex transactions or complex queries
- A medium to high query or write rate – Up to 30,000 IOPS (15,000 reads + 15,000 writes)
- No more than a single worker node or shard
- High durability

- **Do not use Amazon RDS when your application requires:**
- Massive read/write rates (for example, 150,000 write/second)
- Sharding due to high data size or throughput demands
- Simple GET or PUT requests and queries that a NoSQL database can handle
- Relational database management system (RDBMS) customization

Amazon RDS: Clock-hour billing and database characteristics

Clock-hour billing –

- Resources incur charges when running

Database characteristics –

- Physical capacity of database:
  - Engine
  - Size
  - Memory class

Amazon RDS: DB purchase type and multiple DB instances

DB purchase type –

- On-Demand Instances
    - Compute capacity by the hour
- Reserved Instances
    - Low, one-time, upfront payment for database instances that are reserved with a 1-year or 3-year term

Number of DB instances –

- Provision multiple DB instances to handle peak loads

Amazon RDS: Storage

Provisioned storage –

- No charge
  - Backup storage of up to 100 percent of database storage for an active database
- Charge (*GB/month*)
  - Backup storage for terminated DB instances

Additional storage –

- Charge (*GB/month*)
  - Backup storage in addition to provisioned storage

Amazon RDS: Deployment type and data transfer

Requests –

- The number of input and output requests that are made to the database

Deployment type—Storage and I/0 charges vary, depending on whether you deploy to –

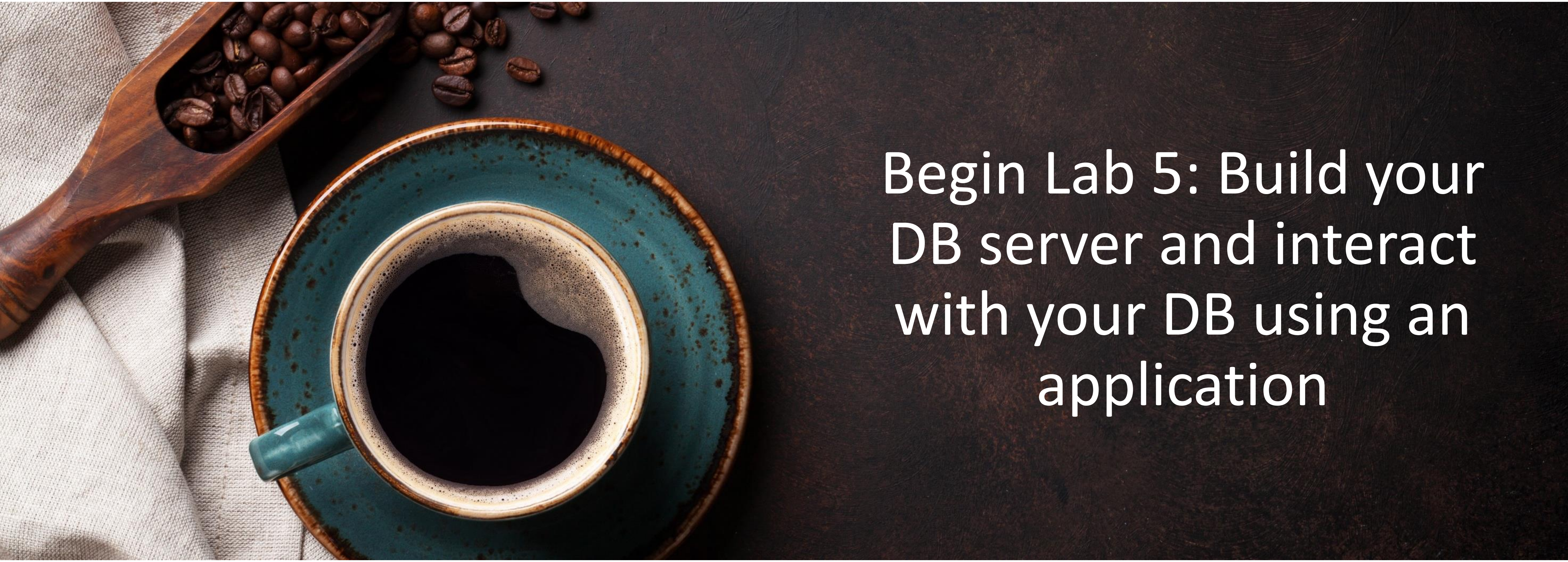
- Single Availability Zone
- Multiple Availability Zones

Data transfer –

- No charge for inbound data transfer
- Tiered charges for outbound data transfer

Recorded demo: Amazon RDS console

~ 30 minutes



Begin Lab 5: Build your DB server and interact with your DB using an application

Section 1 key takeaways

- With Amazon RDS, you can set up, operate, and scale relational databases in the cloud.
- Features –
  - Managed service
  - Accessible via the console, AWS Command Line Interface (AWS CLI), or application programming interface (API) calls
  - Scalable (compute and storage)
  - Automated redundancy and backup are available
  - Supported database engines:
    - Amazon Aurora, PostgreSQL, MySQL, MariaDB, Oracle, Microsoft SQL Server

Section 2: Amazon DynamoDB

Module 8: Databases

Relational versus non-relational databases

<table>
<tr><th></th><th>Relational (SQL)</th><th>Non-Relational</th></tr>
<tr><td>Data Storage</td><td>Rows and columns</td><td>Key-value, document, graph</td></tr>
<tr><td>Schemas</td><td>Fixed</td><td>Dynamic</td></tr>
<tr><td>Querying</td><td>Uses SQL</td><td>Focuses on collection of documents</td></tr>
<tr><td>Scalability</td><td>Vertical</td><td>Horizontal</td></tr>
<tr><td>Example</td><td>
<table>
<tr><th>ISBN</th><th>Title</th><th>Author</th><th>Format</th></tr>
<tr><td>3111111223439</td><td>Withering Depths</td><td>Jackson, Mateo</td><td>Paperback</td></tr>
<tr><td>3122222223439</td><td>Wily Willy</td><td>Wang, Xiulan</td><td>Ebook</td></tr>
</table>
</td><td>{
ISBN: 3111111223439,
Title: “Withering Depths”,
Author: ”Jackson, Mateo”,
Format: “Paperback”
}</td></tr>
</table>

What is Amazon DynamoDB?

Fast and flexible NoSQL database service for any scale

Amazon DynamoDB

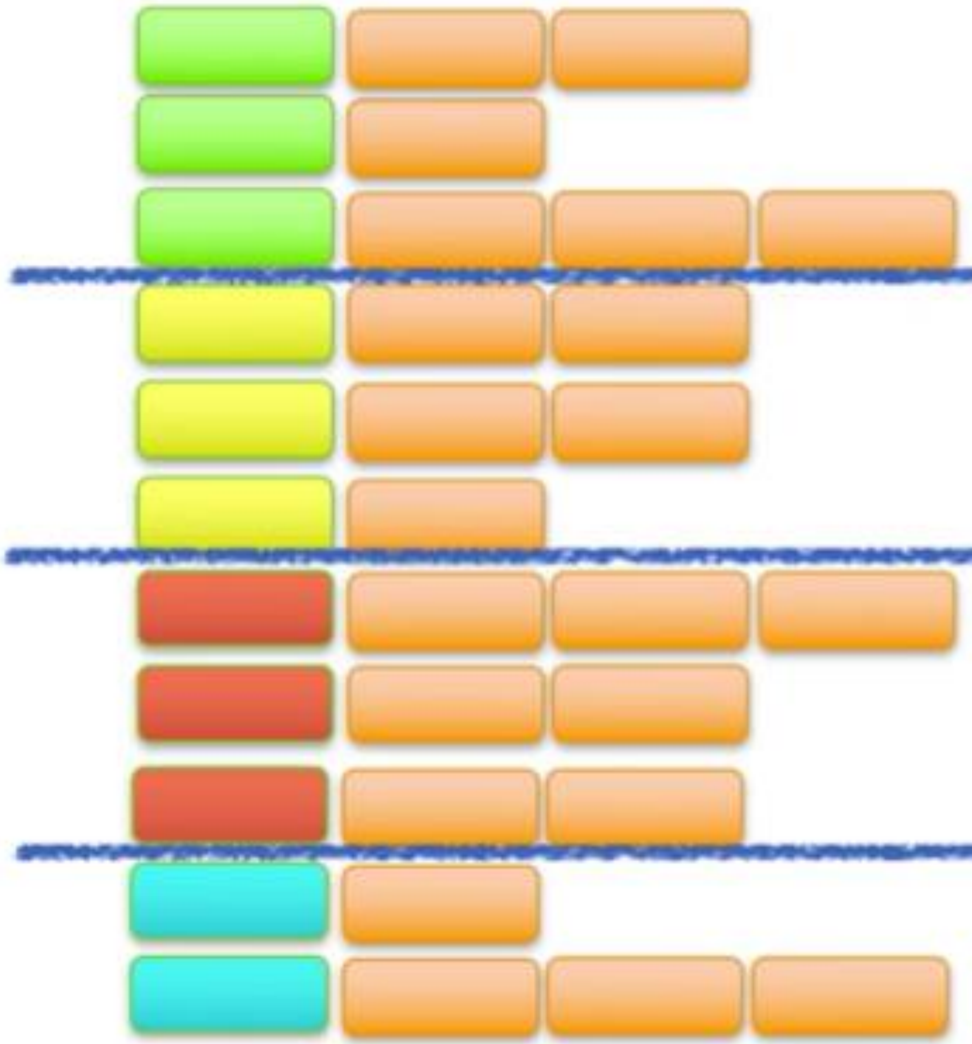
- NoSQL database tables
- Virtually unlimited storage
- Items can have differing attributes
- Low-latency queries
- Scalable read/write throughput

Amazon DynamoDB core components

- Tables, items, and attributes are the core DynamoDB components

- DynamoDB supports two different kinds of primary keys: Partition key and partition and sort key

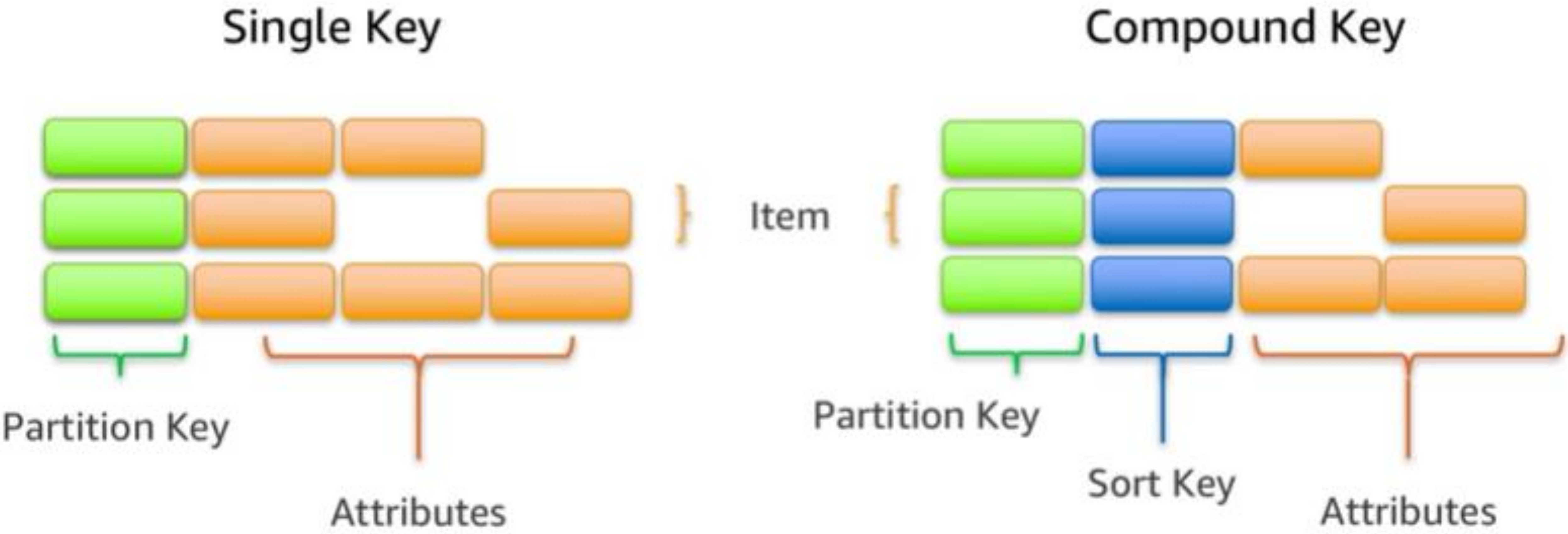
Partitioning



As data grows, table partitioned by key

QUERY by Key to find items efficiently
SCAN to find items by any attribute

Items in a table must have a key



Section 2 key takeaways

Amazon DynamoDB:

- Runs exclusively on SSDs.
- Supports document and key-value store models.
- Replicates your tables automatically across your choice of AWS Regions.
- Works well for mobile, web, gaming, adtech, and Internet of Things (IoT) applications.
- Is accessible via the console, the AWS CLI, and API calls.
- Provides consistent, single-digit millisecond latency at any scale.
- Has no limits on table size or throughput.

Recorded demo: Amazon DynamoDB console

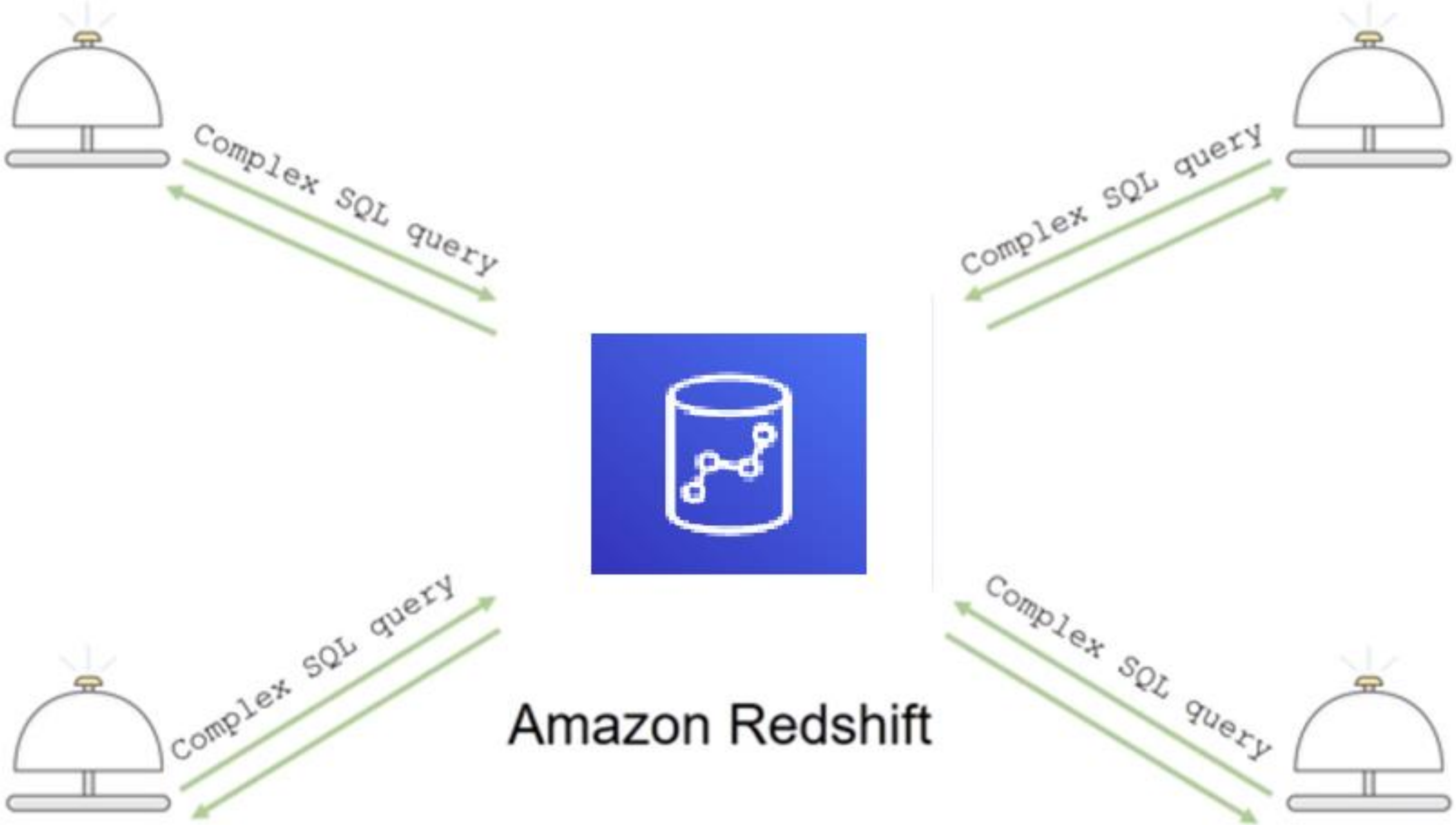
Section 3: Amazon Redshift

Module 8: Databases

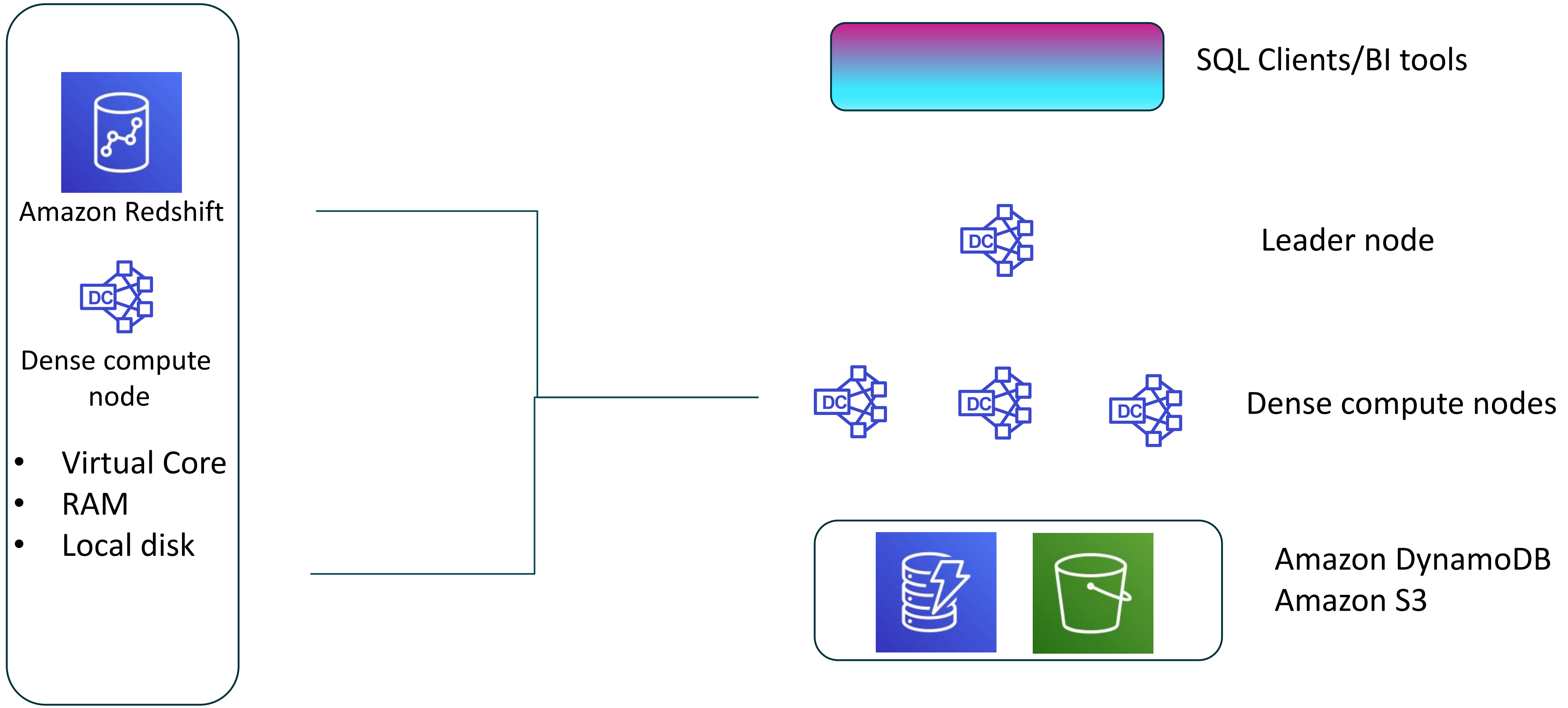
Amazon Redshift

Amazon Redshift

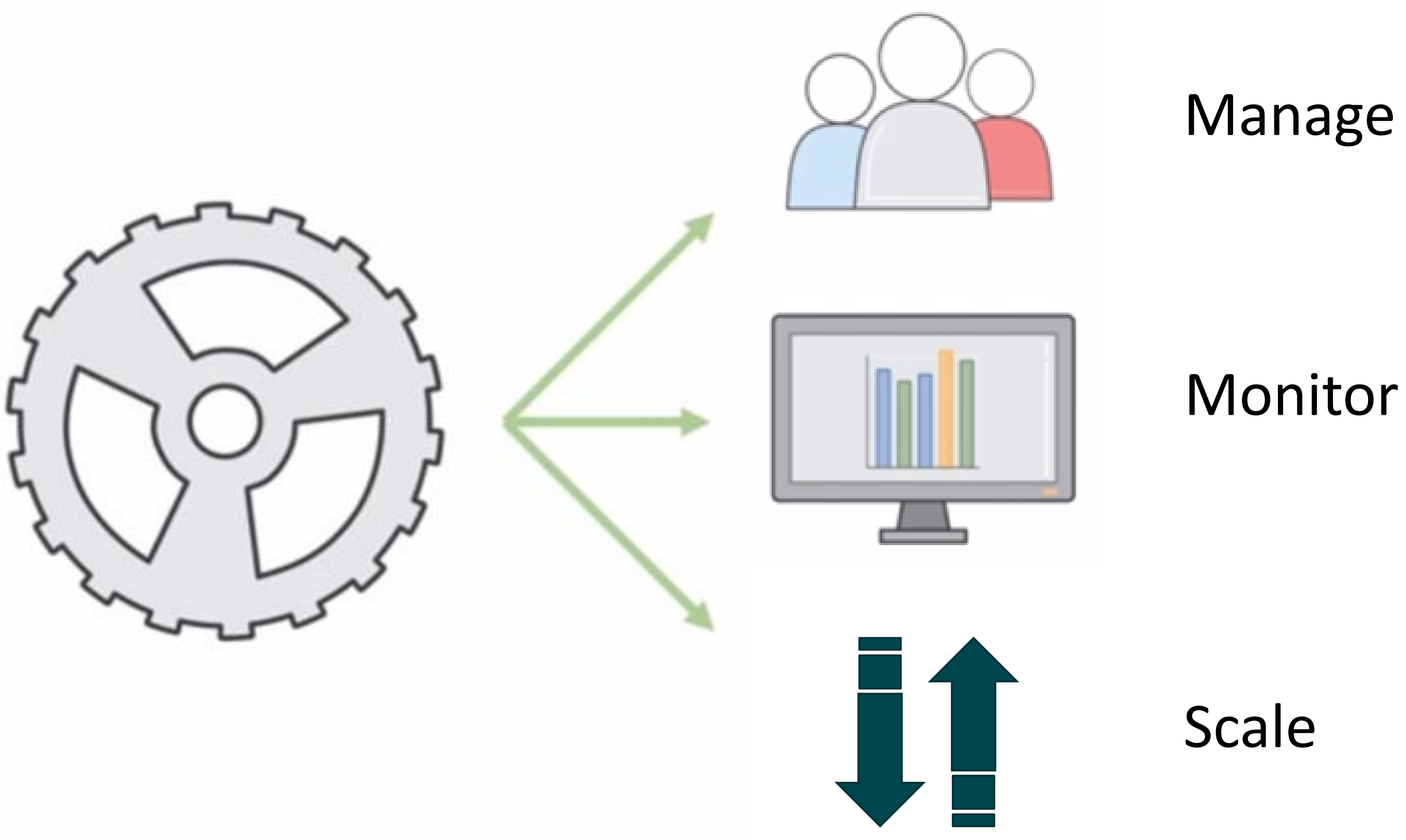
Introduction to Amazon Redshift



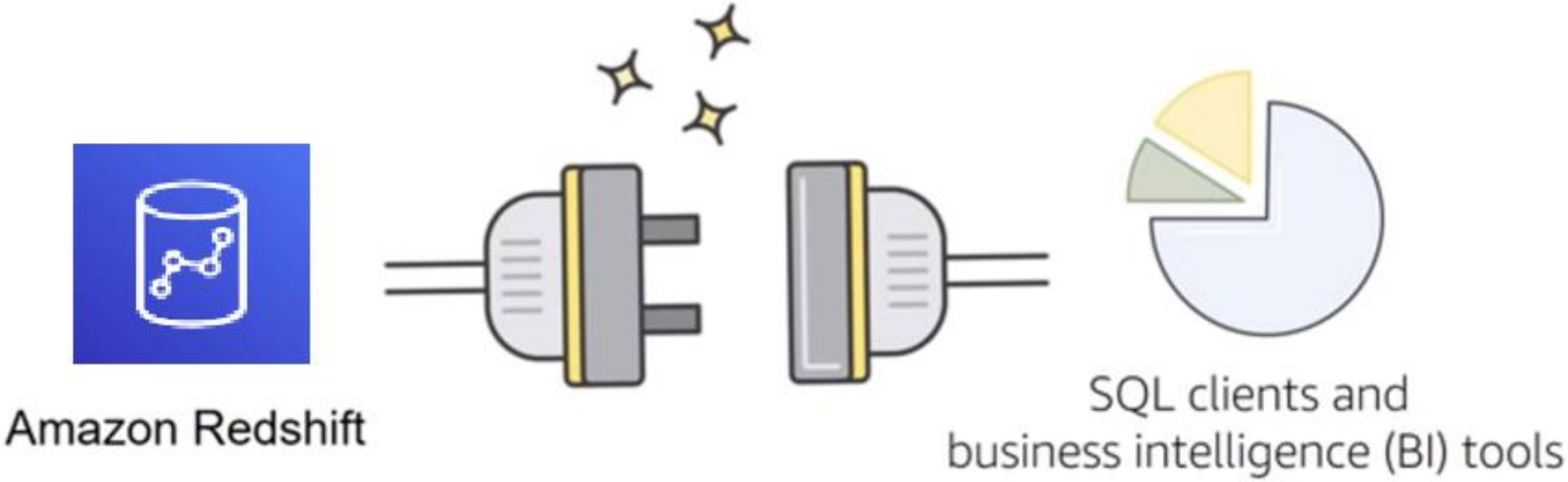
Parallel processing architecture



Automation and scaling

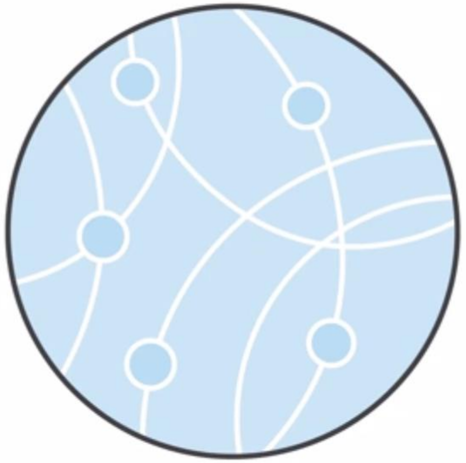


Compatibility



Amazon Redshift use cases (1 of 2)

- Enterprise data warehouse (EDW)
  - Migrate at a pace that customers are comfortable with
  - Experiment without large upfront cost or commitment
  - Respond faster to business needs
- Big data
  - Low price point for small customers
  - Managed service for ease of deployment and maintenance
  - Focus more on data and less on database management



Amazon Redshift use cases (2 of 2)

- Software as a service (SaaS)
  - Scale the data warehouse capacity as demand grows
  - Add analytic functionality to applications
  - Reduce hardware and software costs

Section 3 key takeaways

Amazon Redshift features:

- Fast, fully managed data warehouse service
- Easily scale with no downtime
- Columnar storage and parallel processing architectures
- Automatically and continuously monitors cluster
- Encryption is built in

Section 4: Amazon Aurora

Module 8: Databases

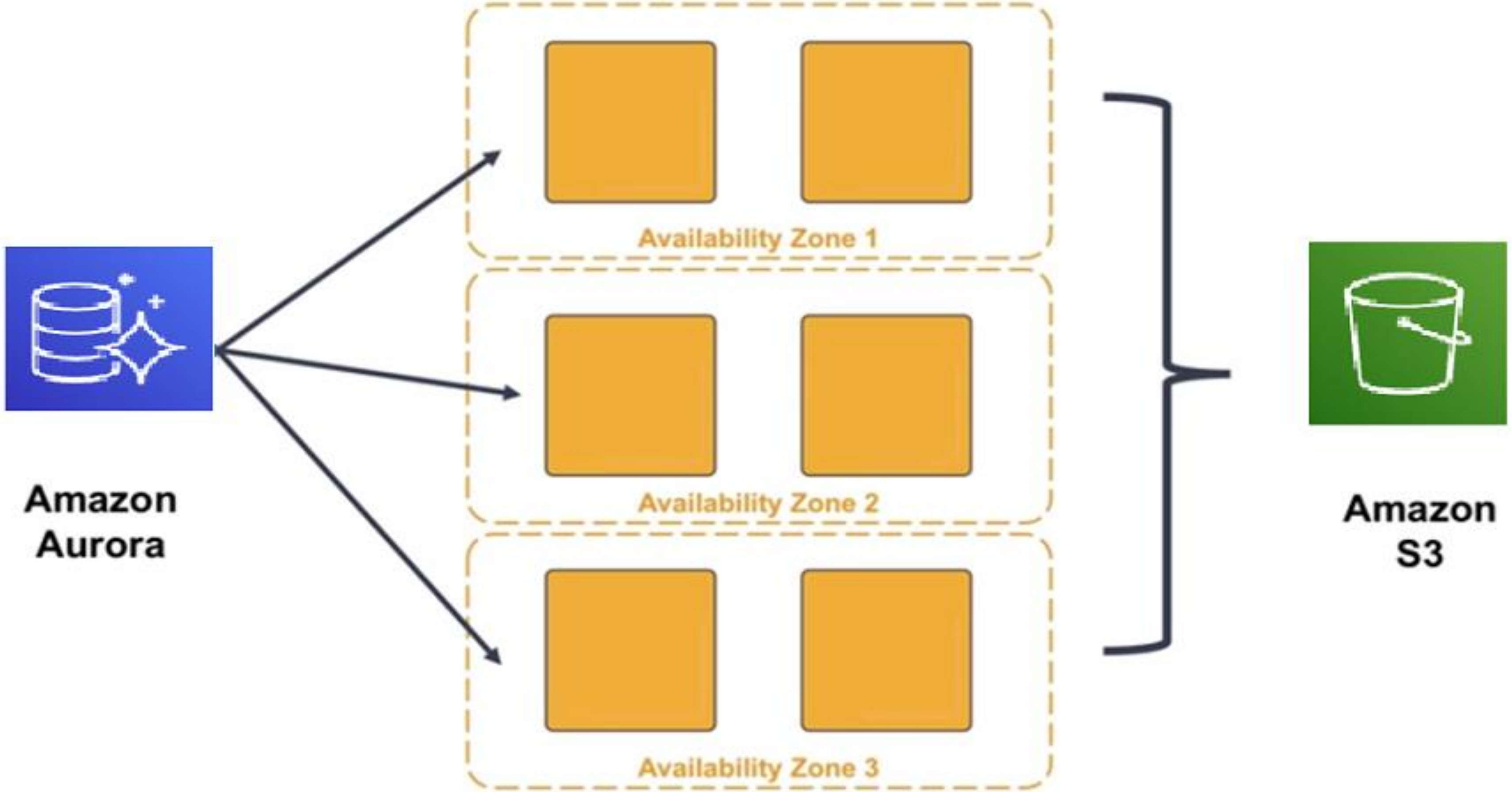
Amazon Aurora

Amazon Aurora

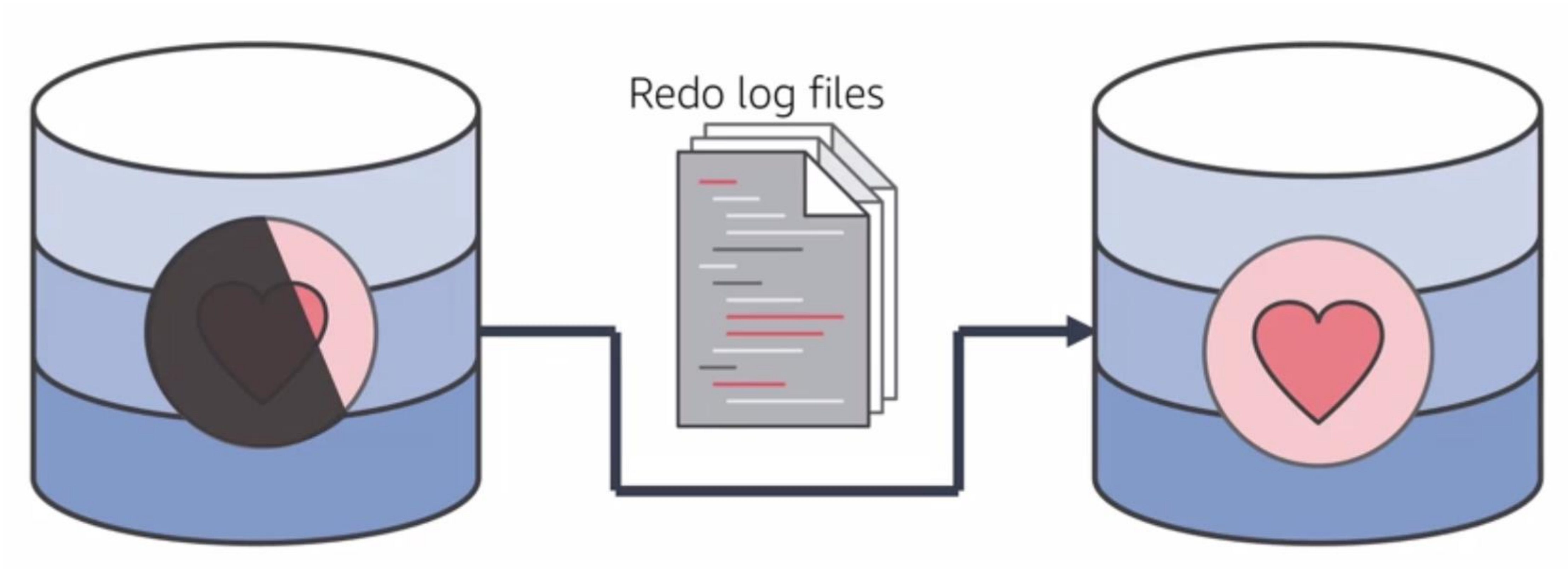
- Enterprise-class relational database
- Compatible with MySQL or PostgreSQL
- Automate time-consuming tasks (such as provisioning, patching, backup, recovery, failure detection, and repair).

Amazon Aurora service benefits

High availability



Resilient design



Section 4 key takeaways

Amazon Aurora features:

- High performance and scalability
- High availability and durability
- Multiple levels of security
- Compatible with MySQL and PostgreSQL
- Fully managed

The right tool for the right job

What are my requirements?	
Enterprise-class relational database	Amazon RDS
Fast and flexible NoSQL database service for any scale	Amazon DynamoDB
Operating system access or application features that are not supported by AWS database services	Databases on Amazon EC2
Specific case-driven requirements (machine learning, data warehouse, graphs)	AWS purpose-built database services

Module wrap-up

Module 8: Databases

Module summary

In summary, in this module, you learned how to:

- Explain Amazon Relational Database Service (Amazon RDS)
- Identify the functionality in Amazon RDS
- Explain Amazon DynamoDB
- Identify the functionality in Amazon DynamoDB
- Explain Amazon Redshift
- Explain Amazon Aurora
- Perform tasks in an RDS database, such as launching, configuring, and interacting

Complete the knowledge check

Sample exam question

Which of the following is a fully-managed NoSQL database service?

Choice	Response
A	Amazon Relational Database Service (Amazon RDS)
B	Amazon DynamoDB
C	Amazon Aurora
D	Amazon Redshift

Additional resources

- AWS Database page: https://aws.amazon.com/products/databases/
- Amazon RDS page: https://aws.amazon.com/rds/
- Overview of Amazon database services: https://docs.aws.amazon.com/whitepapers/latest/aws-overview/database.html
- Getting started with AWS databases: https://aws.amazon.com/products/databases/learn/

Thank you

Corrections, feedback, or other questions?
Contact us at https://support.aws.amazon.com/#/contacts/aws-academy.